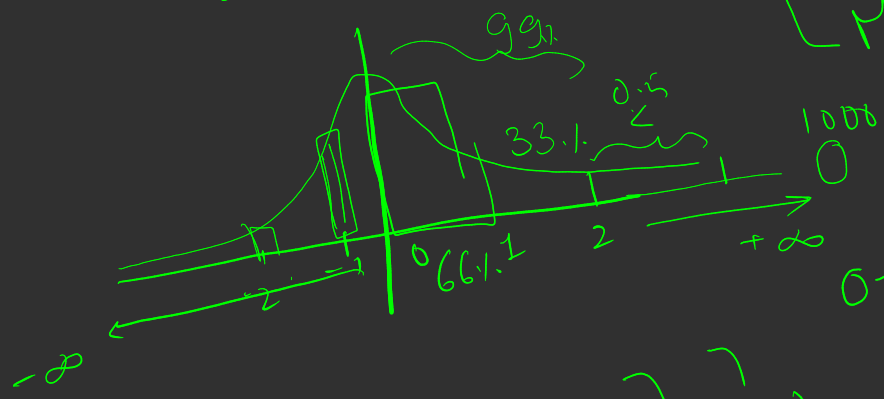
Tutorial on data scaling

Classification
Regression

std ~ $N(0,1)$

MinMax ~ [0, 1]



10^9

0-150 $0-10^9$ USD

2-15

[age, annual income, BMI, pelvic]

$X = (x_1, x_2, x_3, x_4)$

$y = (y_1, y_2, y_3, y_4)$

0-50

$\sqrt{(x_i - y_i)^2}$

Euclidean dist $= \sqrt{\sum (x_i - y_i)^2}$

[age, in, BMI, pelvic]

$X - y$

f

M F

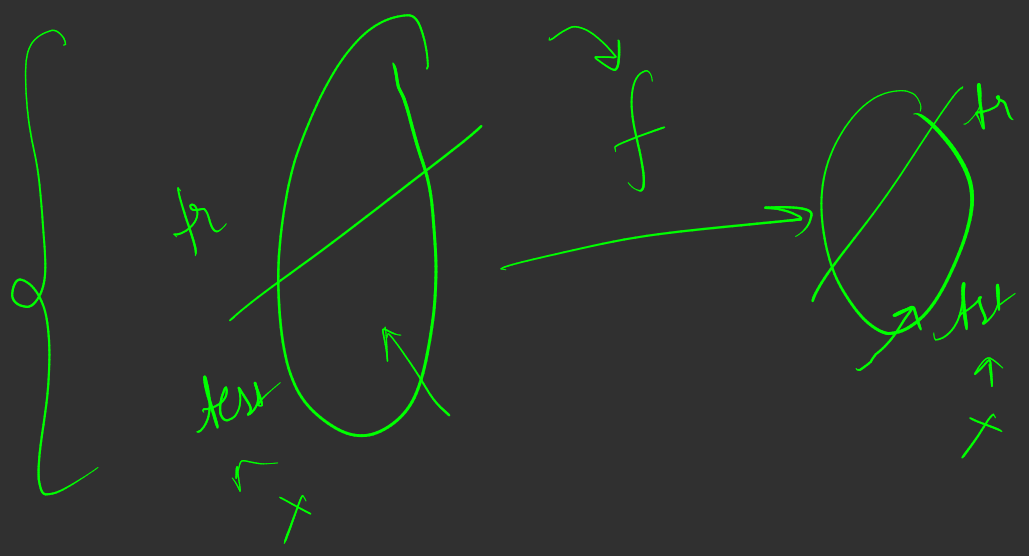
$\frac{M}{F}$

μ

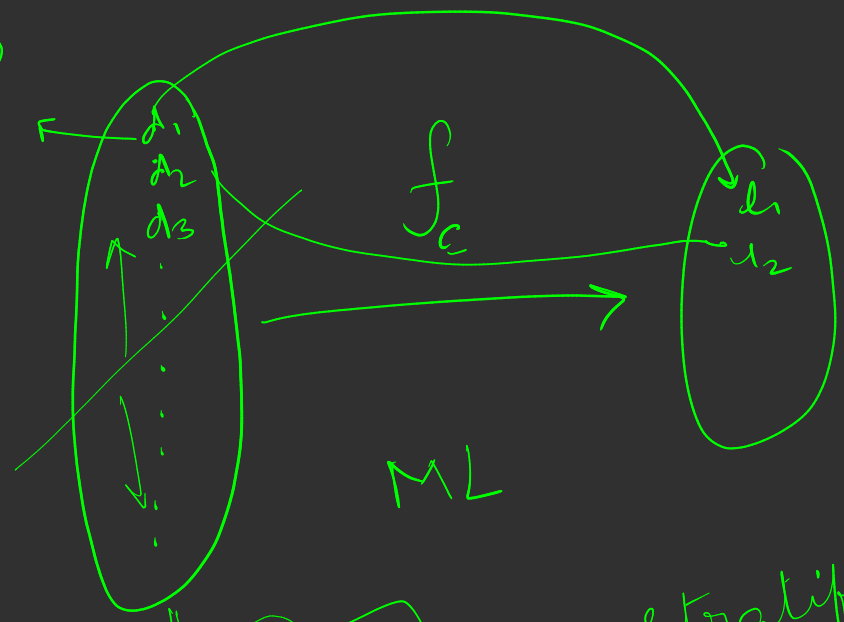
σ

min

max



$(f_1, f_2, f_3, \ldots)$



Supervised learn

ML

Overfit
underfit
Correct fit

stratify

C_1 C_2 C_{10}

C_1 C_2 C_{10}

100 100 100

75 75 75

75 - 25 ✓ 1000

tr 750

250

$N(0,1)$ f_1^1 ..... f_{10} $N(0,1)$

μ_1, σ_1 μ_{10}, σ_{10}

f_1^2

standardization

750

$$\mu_i = \frac{\sum_{n=1}^{750} f_i^n}{N}$$

$$\sigma_i = \sqrt{\frac{\sum (f_i^n - \mu_i)^2}{N}}$$

$$f_i^1 \rightarrow \frac{f_i - \mu_i}{\sigma_i} \sim N(0,1)$$

f_1^{750} ..... f_{10}^{750}

MinMax Scaling

$$m_i = \min\left(f_i^1, f_i^2, \ldots, f_i^{750}\right)$$

$$M_i = \max\left(f_i^1, f_i^2, \ldots, f_i^{750}\right)$$

$$\tilde{f}_i^n = \frac{f_i^n - m_i}{M_i - m_i}$$

$\tilde{f}_i^n \to 0-150$

$\tilde{f}_i^n \sim (0,1)$